# Ajay Venkatesh

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## Skills

**Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL
**AI/ML:** PyTorch, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Hugging Face, SVM, Random Forest, MLflow, PySpark
**Web Development:** REST API, GraphQL API, Microservices, React, Node.js, Express.js
**Database:** RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse
**Cloud:** AWS (AWS CDK, API Gateway, Lambda, IAM, Q, SageMaker), Azure (AI services, Function Apps, Storage blobs)
**Tools:** Git, Visual Studio, Jupyter, Postman, Docker, GitLab CI, Terraform, Ansible, Boto3, Datadog, Power BI

## Experience

**Campus Mesh** **Jan 2026 – Present**
*AI Software Engineer* *New York, US*
- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Designed and implemented a collaborative-filtering **recommendation engine** using user interaction signals, increasing platform engagement by **28%** and user retention by **50%**.
- Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

**Amazon** **May 2025 – Aug 2025**
*Software Development Engineer Intern* *Seattle, US*
- Built a high-throughput, event-driven incident triage system using **AWS ECS Fargate Tasks, SQS, and SNS**, decoupling message processing and reducing incident response latency by **50%** during peak traffic.
- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an agentic workflow that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Improved large-scale log analysis by implementing a **Model Context Protocol (MCP)** server for efficient retrieval, achieving a **40%** reduction in AWS Athena query time and compute costs.
- Provisioned resilient **Infrastructure-as-Code (IaC)** using **AWS CDK** to manage auto-scaling ECS clusters, ensuring **99.9%** availability under 2x projected load.

**New York University (Novel AI Technologies, Inc.)** **Aug 2024 – Present**
*AI Software Engineer* *New York, US*
- Architected an **NSF-funded ($100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented time-series anomaly detection using **Isolation Forest** and **LSTM** models.
- Built **computer vision pipelines (YOLOv8, EfficientNet)** to process user-submitted food images integrated with **GPT-4 workflows** for nutritional estimation and dietary feedback.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)** integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
- Engineered ETL pipelines transforming **NoSQL** data into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.

**PricewaterhouseCoopers (PwC)** **Aug 2021 – Aug 2024**
*Senior Software Engineer*
- Designed and implemented a distributed backend system on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting ML inference pipelines for healthcare analytics.
- Led code reviews and mentored interns on **software design and test-driven development**, reducing production bugs by **25%**.
- Built an **NLP-powered OCR pipeline** using **LayoutLM** and **Azure Form Recognizer** to automate data extraction from medical documents, reducing processing time by **75%**.
- Optimized data ingestion using **parallel processing** for **100K+** Excel records, reducing execution time from **5 hours to 1 hour**.
- Developed **Power BI dashboards** integrating multiple enterprise databases, improving reporting efficiency by **40%**.

## Projects

**InviLite (AI-Driven Data Engineering & Analytics Platform)** | ***AWS, SageMaker, Bedrock, FAISS, Python***
- Architected a cloud-native data infrastructure (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a **RAG system** (SVM classification + FAISS vector indexing + Bedrock LLM) enabling contextual resolution recommendations.
- Reduced incident resolution time from **1–2 hours to minutes** and improved historical match accuracy by **30%**.
- Increased root-cause classification accuracy from **56% to 92%** through structured feature engineering and model retraining.
- Implemented dual-layer (real-time + batch) Lambda architecture supporting continuous learning and inference scalability.

## Education

**New York University** | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Aug 2024 – May 2026

## Certifications & Publications

**AWS Cloud Practitioner**
Stanford Machine Learning (Coursera)
Power BI Data Analyst Associate
Azure Data Fundamentals

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*
An Aquila-optimized SVM classifier for Diabetes prediction *(IEEE)*